

Lab 1: Requirements Engineering & UML Use-Case Modelling

Scenario: Food Bank Surplus Redistribution Platform

Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow donors to post surplus food packages with perishable expiry countdown timers and dietary tags.	High	Pass: Shelter reserves batch and receives pickup verification code; Fail: Expired food batch remains active on claim board.	Allows donors to make surplus food available to shelters before it expires.
FR-002	Functional	The system shall allow verified shelters to reserve a pickup window for an available food batch before its expiry.	High	When a verified shelter selects an available food batch and chooses an available pickup window before expiry, the system records the reservation and displays a confirmation.	Enables shelters to claim available surplus food and arrange collection before expiry.
FR-003	Functional	The system shall allow verified shelters to view available surplus food batches based on location and expiry time	High	When a verified shelter searches for available food, the system displays eligible batches with their location and expiry information.	Helps shelters identify suitable food batches before they expire.
FR-004	Functional	The system shall notify verified shelters within a 5 km radius when high-quality perishable food is listed.	High	When a qualifying food batch is posted, verified shelters within 5 km receive a notification containing the available batch details.	Helps nearby shelters respond quickly to perishable food availability
FR-005	Functional	The system shall provide a pickup verification code for a reserved food batch.	High	After a shelter successfully reserves a food batch, the system generates and displays a unique pickup verification code.	Provides a way to verify the authorized pickup of the reserved food batch.
NFR-001	Nonfunctional	The platform shall dispatch instant push notifications to shelters within a 5 km radius when high-quality perishable food is listed.	High	Pass: Benchmarking tests confirm the target notification latency and security standards under simulated peak load.	Ensures nearby shelters are informed quickly enough to act before perishable food expires.
NFR-002	Nonfunctional	The platform shall prevent unauthorized users from accessing donor and shelter account information.	High	Unauthorized access attempts to protected account information are denied in all security test cases	Protects sensitive information belonging to donors and shelters.